

CHANGE REQUEST

LoRaWAN Link Layer specification	version	1.0.4	Reference doc. ID	
<i>causeway doc nb</i>				
CR Title:	ERRATA Allow Downlink Transmissions		Revision	1
<i>starts from 1</i>				
Source:	Senet, Inc. (Dave Kjendal – dkjendal@senetco.com)			
Topic or Working Group:	TC	Date:	2022-10-17	
Category:	E E (errata) F (correction) A (corresponds to a correction in an earlier release)			
	B (feature addition) C (functional modification of a feature) D (editorial modification / clarification)			
Reason for change:	Remove the mistaken prohibition of end-devices from sending downlink frames			
Summary of change:	Removal of prohibition			
Clauses affected:	3.2 Downlink Frames 4.3.1.5 Frame Counter (FCnt) 9.3 Downlink Ping Frames			
Affected clauses with impact on certification test:	None			
<i>please note whether clause modification leads to (new) or (modified) test</i>				

1. Discussion

The description of downlink is overly restrictive, but not normative. Clarify to remove the restriction which imply that D2D is prohibited (same text as 1.2.0). Relax other prohibitions.

2. Proposed changes (normative/informative)

***** Start of proposed change for 3.2 *****

3.2 Downlink Messages

~~Downlink packets are sent to one or more end-devices, typically transmitted by a Network Server through one or more gateways. Each downlink message is sent by the Network Server to only one end device and is relayed by a single gateway.³~~

Commented [DK1]: Remove footnote, it is incorrect (says we don't describe multicast, but we do)

***** End of proposed change for 3.2 *****

4.3.1.5 Frame counter (FCnt)

There are two frame counters for each end-device. FCntUp is incremented by an end-device when a data frame is transmitted to a Network Server (uplink). FCntDown is incremented ~~by a Network Server~~ when a data frame is transmitted to an end-device (downlink). ~~The Network Server tracks the uplink frame counter and generates the downlink counter for each end-device.~~

Whenever an OTAA end-device successfully processes a Join-Accept frame, the frame counters on the end-device (FCntUp) and the Network side (FCntDown) are reset to 0.

For ABP (activation by personalization) end-devices, the frame counters are initialized to 0 by the manufacturer. ABP end-devices SHALL NOT reset the frame counters during the end-device's lifetime. If the end-device is susceptible to losing power during its lifetime (battery replacement, for example), the frame counters SHALL persist during such an event.

Subsequently, FCntUp is incremented with each uplink and FCntDown is incremented with each downlink. At the receiver side, the corresponding counter is kept in sync with the received value, provided the received value has been incremented compared to the current counter value, and the frame MIC field matches the MIC value computed locally using the appropriate network session key. FCntUp SHALL NOT be incremented in the case of multiple transmissions of a confirmed or unconfirmed frame (see NbTrans parameter). Network Servers SHALL drop the application payload of the retransmitted frames and only forward a single instance to the appropriate Application Server.

A first uplink with FCntUp=0 sent by an ABP or an OTAA end-device after a successful Join procedure SHALL be accepted by a Network Server, provided the MIC field is valid.

³ ~~This specification does not describe the transmission of multicast messages from a network server to many end-devices.~~



Analogously, a first downlink with `FCntDown=0` sent by a Network Server to an ABP or an OTAA end-device after a successful Join procedure SHALL be accepted by the end-device, provided the `MIC` field is valid.

Frame counters are 32 bits wide. The `FCnt` field SHALL correspond to the least-significant 16 bits of the 32-bit frame counter (i.e., `FCntUp` for data frames sent uplink and `FCntDown` for data frames sent downlink).

The end-device SHALL NOT reuse the same `FCntUp` value with the same application or network session keys, except for retransmission.

The end-device SHALL NOT process any retransmission of the same downlink frame.

Subsequent retransmissions SHALL be ignored without being processed.

***** *Start of proposed change for 9.3* *****

9.3 Downlink Ping Frames

A downlink ping frame uses the same format as a Class A downlink frame but might follow a different channel frequency or data rate plan.

Frames can be unicast or multicast. Unicast frames are sent to a single end-device, whereas multicast frames are sent to multiple end-devices. All end-devices of a multicast group SHALL share the same multicast address and associated encryption keys. A Class B-capable end-device SHALL support at least one multicast-group ~~and an end-device SHALL NOT transmit using a multicast address.~~

***** *End of proposed change for 9.3* *****